EXPERIMENT – 1 **DATE: 22-10-2026**

ARDUINO UNO-BASED SMART HOME SAFETY AND OCCUPANCY MONITORING SYSTEM USING GAS, PIR AND ULTRASONIC SENSORS

Objectives of this Experiment:

1. To understand the interfacing of gas, PIR and ultrasonic sensors with the Arduino UNO.
2. To acquire an analog gas sensor value using the ADC of the Arduino UNO.
3. To detect human motion/occupancy using a PIR sensor.
4. To measure object distance using an ultrasonic sensor.
5. To implement safety and occupancy decision logic using Embedded C programming.
6. To provide visual and audible alerts using an LED and buzzer.
7. To display real-time sensor information using the Proteus Virtual Terminal.
8. To simulate and verify the complete smart home safety monitoring system using Proteus.

OUTPUT PARAMETER TABLE – VALUES TO BE RECORDED

S. No.	Output Parameter	Expected Value / Condition	Practical / Simulated Observation
1	Gas Level	ADC value displayed	Verified
2	Normal Gas Condition	Gas ≤ 500	Verified
3	Gas Alert	Gas > 500	Verified
4	PIR Status	Detected / Not Detected	Verified
5	Occupancy Detection	Occupancy Detected	Verified
6	Ultrasonic Distance	Distance in cm	Verified
7	LED Operation	ON during motion	Verified
8	Buzzer Operation	ON during gas alert	Verified
9	Safe Condition	System Safe	Verified
10	Serial Monitoring	Continuous sensor display	Verified

RUBRIC:

Objective	Assessment Criterion	Weightage	Marks
Develop the Proteus circuit	Circuit Design and Configuration	2.5	
Develop the Embedded C program	Program Development	2.5	
Implement sensor processing	Sensor Acquisition and Decision Logic	2.5	
Verify system operation	Simulation and Result Analysis	2.5	
Total		**10**	

Experiment uploaded in GitHub on :

https://github.com/madurivadapalli-tech/192311368.EMBEDDED/blob/main/LAB%20EXPERIMENTS/EXPERIMENT%201.11368.pdf